Group Number: Team 8 **Group Members:** Mikail & Kyle

Review: ***[Boundary Analysis]*** Team 8 did a strong job identifying a wide range of rules in alignment with the specifications outlined in the GE2 Statement. However, some differences emerged when comparing their analysis to ours, particularly in how ranges of values were handled.

For example, the project_acronym variable is defined as having a minimum length of 5 characters and a maximum of 10 characters. While this is correctly reflected in their valid test cases, the distinction between boundary values and equivalence class values is not clearly or consistently presented. As discussed in class, value ranges should be explicitly defined (e.g. 5 ≤ length ≤ 10), with boundary values identified at the limits (5 and 10), as well as representative values within the range (e.g., 6 and 9). We noticed that many of the boundary values are defined as ranged, rather than having concrete testable values, which makes it harder to translate to test cases.

One element we noticed during the cross-review was that team 8 grouped many of the rules together for each variable. For example, datatypes and syntax rules were combined under a single rule description. Additionally, there were no mentions of Datatype as an ECV altogether, which could present errors in the future if incorrect datatypes are passed into the function. It is important to test this to ensure the reliability of the software and to ensure that these issues are caught early in the development process.

Another recommendation is to further clarify ambiguous criteria such as “free text” in the project description section. In our approach, we used regular expressions (regex) to provide a more precise and unambiguous definition, reducing the likelihood of misinterpretation by future readers.

Overall, Team 8 demonstrated a thorough evaluation of the criteria defined in the GE2 statement. While there are some minor formatting and clarity improvements that could be made, their general analysis matches our interpretation.

[Test Cases] Team 8 did a strong job translating their boundary value analysis and equivalence class definitions into concrete test cases that can effectively evaluate the correctness of their program.

One aspect we found particularly interesting was their inclusion of a system_state variable. While this was not explicitly defined in the GE2 statement as a parameter to be tested, incorporating it provides a more holistic view of the system. That said, it may not have been strictly necessary within the scope of Boundary Value and Equivalence Class analysis.

One recommendation would be to include more descriptive context in their exception messages. Providing clearer error details in the error messages would make it easier to trace issues back to their root causes and simplify debugging in later stages of development.

Another point of critique is the use of the “(valid fillers)” notation in the ECs and BVs Covered section. While this abstraction is understandable from a development efficiency standpoint, it makes it more difficult to cross-reference test cases with their corresponding equivalence classes and boundary values. The approach itself is thoughtful, but adding a clearer definition of what these fillers represent would improve readability and accessibility for someone reviewing the work without prior context.

Overall, the approach is well-structured and demonstrates strong consideration of testing principles, with a few opportunities to improve clarity and traceability.

Boundary Analysis:

VARIABLE	Rule	VALID CLASSES	INVALID CLASSSES
company_cif	Tax Identification Code of the company. -> Implementation of the CIF validation algorithm (Letter + 7 digits + Control).	**ECV1: Valid CIF with letter control.**	**ECNV1: Invalid control character.**
		ECV2: Valid CIF with number control.	**BVNV1: Length < 9 chars.**
			BVNV2: Length > 9 chars.
			ECNV2: Incorrect format (e.g., missing starting letter).
project_achronym	Maximum 10 characters, minimum 5. Characters allowed: A-Z or numbers (0-9).	**BVV1: Exactly 5 characters.**	**BVNV3: Exactly 4 characters.**
		BVV2: Exactly 10 characters.	**BVNV4: Exactly 11 characters.**
		ECV3: Between 6 and 9 characters.	**ECNV3: Contains lowercase letters (a-z).**
		ECV4: Only contains A-Z.	**ECNV4: Contains special characters (e.g., -, _, @).**
		ECV5: Only contains 0-9.	
project_description	Project name (free text between 10 and 30 characters).	**BVV3: Exactly 10 characters.**	**BVNV5: Exactly 9 characters.**
		BVV4: Exactly 30 characters.	**BVNV6: Exactly 31 characters.**
		ECV6: Between 11 and 29 characters.	**ECNV5: Empty string (0 characters).**
department	Responsible department. Possible values: HR, FINANCE, LEGAL, LOGISTICS.	**ECV7: HR**	**ECNV6: Value not in the allowed list (e.g., IT, SALES).**
		ECV8: FINANCE	**ECNV7: Lowercase or mixed case (e.g., Finance, hr).**

		ECV9: LEGAL	**ECNV8: Empty string.**
		ECV10: LOGISTICS	
date	It must be in the format "DD/MM/YYYY". A valid date between the year 2025 and 2027. The date must be equal to or after the date on which the request to register the project was received.	**ECV11: Valid format "DD/MM/YYYY".**	**BVNV7: Year is 2024.**
		BVV5: Year is 2025.	**BVNV8: Year is 2028.**
		BVV6: Year is 2027.	**ECNV9: Date is before the request date.**
		ECV12: Year is 2026.	**ECNV10: Invalid format (e.g., "YYYY-MM-DD", missing slashes).**
		BVV7: Date is exactly the request date.	**ECNV11: Invalid calendar date (e.g., "32/01/2026").**
		ECV13: Date is in the future.	
budget	The parameter will receive a number between 50000.00 and 1000000.00. The value received will always have 2 decimal places.	**BVV8: Exactly 50000.00.**	**BVNV9: Exactly 49999.99.**
		BVV9: Exactly 1000000.00.	**BVNV10: Exactly 1000000.01.**
		ECV14: Between 50000.01 and 999999.99.	**ECNV12: Incorrect decimal places (e.g., 50000.1, 50000.000).**
			ECNV13: Non-numeric characters (if applicable).
OUTPUT	If inputs are valid, generate MD5 ID and store in JSON. EnterpriseManagementException is raised if inputs are invalid or a project with the same name for the same CIF already exists in the data file.	**ECV15: Project with this acronym and CIF does not exist yet in corporate_operations.json. (Returns MD5 string and saves data) .**	**ECNV14: Inputs are invalid or project with the same acronym for the same CIF already exists in corporate_operations.json. (Raises Exception) .**

Test Cases:

#	I/O	TECHNIQUE	VALID/INVALID	ID TEST	DESCRIPTION	company_cif	project_achronym	project_description	department	date	budget	system_state	RESULT	ECS, BV COVERED
1	Input/Output	EC	Valid	TC01	Base valid case (Happy Path) combining standard ECs.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Valid, String in MD5 Format	ECV2, ECV3, ECV6, ECV8, ECV11, ECV12, ECV13, ECV14, **ECV15**
2	Input/Output	BVA	Valid	TC02	Minimum boundary values for all possible fields.	"P1234567J"	"ABCDE"	"TenChars__"	"HR"	"26/02/2026"	50000.00	Does not exist	Valid, String in MD5 Format	ECV1, BVV1, ECV4, BVV3, ECV7, BVV5, BVV7, BVV8, **ECV15**
3	Input/Output	BVA	Valid	TC03	Maximum boundary values for all possible fields.	"B12345674"	"0123456789"	"ThirtyCharsThirtyCharsThirtyCh"	"LEGAL"	"31/12/2027"	1000000.00	Does not exist	Valid, String in MD5 Format	BVV2, ECV5, BVV4, ECV9, BVV6, BVV9, **ECV15**
4	Input/Output	EC	Valid	TC04	Covers remaining Valid ECs (Mixed Acronym, Logistics).	"A12345674"	"PRJ01M"	"Valid Project Desc"	"LOGISTICS"	"15/05/2026"	75000.00	Does not exist	Valid, String in MD5 Format	ECV10, **ECV15**
5	Input	EC	Invalid	TC05	Invalid CIF: Incorrect control character calculation.	"A12345675"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**ECNV1, (Valid fillers)**

6	Input	EC	Invalid	TC06	Invalid CIF: Incorrect format (missing starting letter).	"123456789"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**ECNV2, (Valid fillers)**
7	Input	BVA	Invalid	TC07	Invalid CIF: Length is too short (< 9 chars).	"A1234567"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**BVNV1, (Valid fillers)**
8	Input	BVA	Invalid	TC08	Invalid CIF: Length is too long (> 9 chars).	"A12345674X"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**BVNV2, (Valid fillers)**
9	Input	BVA	Invalid	TC09	Invalid Acronym: Length is too short (< 5 chars).	"A12345674"	"PRJ1"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**BVNV3, (Valid fillers)**
10	Input	BVA	Invalid	TC10	Invalid Acronym: Length is too long (> 10 chars).	"A12345674"	"PRJ01234567"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**BVNV4, (Valid fillers)**
11	Input	EC	Invalid	TC11	Invalid Acronym: Contains lowercase letters.	"A12345674"	"prj01"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**ECNV3, (Valid fillers)**
12	Input	EC	Invalid	TC12	Invalid Acronym: Contains special characters.	"A12345674"	"PRJ-01"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**ECNV4, (Valid fillers)**
13	Input	BVA	Invalid	TC13	Invalid Description: Length is too short (< 10 chars).	"A12345674"	"PRJ012"	"TooShort!"	"FINANCE"	"15/05/2026"	75000.00	Does not exist	Exception	**BVNV5, (Valid fillers)**

14	Input	BVA	Invali d	TC14	Invalid Description: Length is too long (> 30 chars).	"A1234 5674"	"PRJ0 12"	"ThisDescri ptionIsWay ToooooLo ng"	"FINA NCE"	"15/05 /2026"	7500 0.00	Does not exist	Exception	**BVNV6, (Valid fillers)**
15	Input	EC	Invali d	TC15	Invalid Description: Empty string.	"A1234 5674"	"PRJ0 12"	""	"FINA NCE"	"15/05 /2026"	7500 0.00	Does not exist	Exception	**ECNV5, (Valid fillers)**
16	Input	EC	Invali d	TC16	Invalid Department: Value not in the allowed list.	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	"IT"	"15/05 /2026"	7500 0.00	Does not exist	Exception	**ECNV6, (Valid fillers)**
17	Input	EC	Invali d	TC17	Invalid Department: Lowercase or mixed case.	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	"Fina nce"	"15/05 /2026"	7500 0.00	Does not exist	Exception	**ECNV7, (Valid fillers)**
18	Input	EC	Invali d	TC18	Invalid Department: Empty string.	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	""	"15/05 /2026"	7500 0.00	Does not exist	Exception	**ECNV8, (Valid fillers)**
19	Input	BVA	Invali d	TC19	Invalid Date: Year is 2024 (Below minimum).	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	"FINA NCE"	"31/12 /2024"	7500 0.00	Does not exist	Exception	**BVNV7, (Valid fillers)**
20	Input	BVA	Invali d	TC20	Invalid Date: Year is 2028 (Above maximum).	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	"FINA NCE"	"01/01 /2028"	7500 0.00	Does not exist	Exception	**BVNV8, (Valid fillers)**
21	Input	EC	Invali d	TC21	Invalid Date: Date is before the request date.	"A1234 5674"	"PRJ0 12"	"Valid Project Desc"	"FINA NCE"	"25/02 /2026"	7500 0.00	Does not exist	Exception	**ECNV9, (Valid fillers)**

22	Input	EC	Invalid	TC22	Invalid Date: Invalid format (YYYY-MM-DD).	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"2026-05-15"	75000.00	Does not exist	Exception	**ECNV10, (Valid fillers)**
23	Input	EC	Invalid	TC23	Invalid Date: Invalid calendar date (e.g. 32nd day).	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"32/01/2026"	75000.00	Does not exist	Exception	**ECNV11, (Valid fillers)**
24	Input	BVA	Invalid	TC24	Invalid Budget: Just below minimum boundary.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	49999.99	Does not exist	Exception	**BVNV9, (Valid fillers)**
25	Input	BVA	Invalid	TC25	Invalid Budget: Just above maximum boundary.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	1000000.01	Does not exist	Exception	**BVNV10, (Valid fillers)**
26	Input	EC	Invalid	TC26	Invalid Budget: Incorrect decimal places.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.1	Does not exist	Exception	**ECNV12, (Valid fillers)**
27	Input	EC	Invalid	TC27	Invalid Budget: Non-numeric / String.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	"75k"	Does not exist	Exception	**ECNV13, (Valid fillers)**
28	Output	EC	Invalid	TC28	Process Error: Project with same Acronym for this CIF already exists.	"A12345674"	"PRJ012"	"Valid Project Desc"	"FINANCE"	"15/05/2026"	75000.00	Already exists	Exception	**ECNV14, (Valid fillers)**